

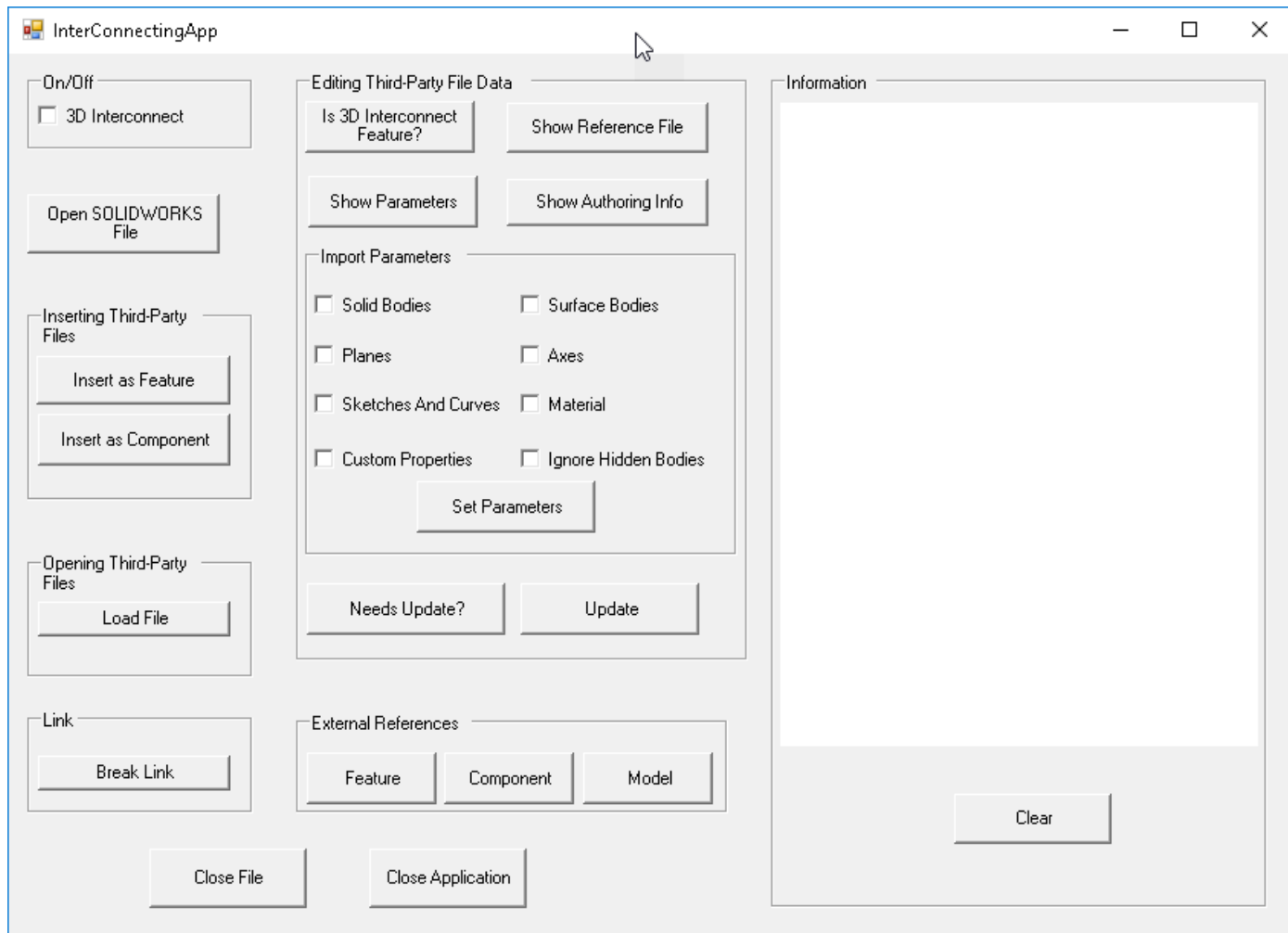
SOLIDWORKS API (.NET VBA C#) — Examples

(36/50): Import3DInterconnect_Example_CSharp (part 1 of 1)

SOLIDWORKS API Documentation (converted from CHM)

3D Interconnect to Import Third-Party Native CAD Files Example (C#)

This example demonstrates how to use the 3D Interconnect API to import and access third-party non-native CAD data.



```
//-----
// Preconditions:
// 1. Ensure Tools > Options > System Options > Enable VSTA VERSION 3.0 is
//    selected.
// 2. Open a new C# macro in SOLIDWORKS Design. Save it with project name
//    Interconnect to a convenient location on your machine.
// 3. Add references, SolidWorks.interop.sldworks and
//    SolidWorks.Interop.swconst.
// 4. Open SolidWorksMacro.cs.
// 5. Replace everything in SolidWorksMacro.cs with this code.
// 6. Right-click the Interconnect project in Solution Explorer and select
//    Add > Windows Form.
// 7. Save the form with name, InterConnectingApp.cs.
// 8. Open InterConnectingApp.Designer.cs and replace everything in it
//    with this code.
// 9. Open the InterConnectingApp.cs form. Right-click on the design
//    and select View Code.
// 10. Replace everything in InterConnectingApp.cs with this code.
// 11. Compile and save the project.
//
// Postconditions:
```

```
// 1. Opens the InterConnectingApp dialog.
// 2. Select 3D Interconnect checkbox to enable 3D Interconnect.
// 3. Click one of:
//     a. Load File
//     b. Open SOLIDWORKS File
// 4. If you clicked 3b, then click one of:
//     a. Insert as Feature (if you opened a part)
//     b. Insert as Component (if you opened an assembly)
// 5. Select the third-party non-native CAD file to insert. Click OK.
// 6. Select the top-level imported item in the FeatureManager design tree.
// 7. Exercise the buttons in the Editing Third-Party File Data group box.
//     Observe the Information text box. Click Clear to refresh it.
// 8. Click Break Link to break the link between the imported
//     feature/component and its original file.
// 9. Click Feature, Component, or Model to list all external references.
// 10. Click Close File to close the current files.
// 11. Click Close Application to close the InterConnectingApp dialog.
//
// -----
```

```
//SolidWorksMacro.cs:
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows;
using System.Windows.Forms;
```

```
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;
```

```
namespace Interconnect
```

```
{
    public partial class SolidWorksMacro
    {
        public void Main()
        {
```

```
swApp.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swStopDebuggingVstaOnExit, false);
        InterConnectingApp obj = new InterConnectingApp();
        obj.RunInterconnectApplication(swApp);
        return;
    }
```

```
    // The SldWorks swApp variable is pre-assigned for you.
```

```
    public SldWorks swApp;
```

```
}
```

```
}
```

```
//InterConnectingApp.Designer.cs:
```

```
namespace Interconnect
```

```
{
```

```
    partial class InterConnectingApp
```

```
    {
```

```
        /// <summary>
```

```
        /// Required designer variable.
```

```
        /// </summary>
```

```
        private System.ComponentModel.IContainer components = null;
```

```
        /// <summary>
```

```
        /// Clean up any resources being used.
```

```
        /// </summary>
```

```
        /// <param name="disposing">true if managed resources should be disposed; otherwise,  
false.</param>
```

```
        protected override void Dispose(bool disposing)
```

```
        {
```

```
            if (disposing && (components != null))
```

```
            {
```

```
                components.Dispose();
```

```
            }
```

```
            base.Dispose(disposing);
```

```
        }
```

```
#region Windows Form Designer generated code
```

```
    /// <summary>
```

```
    /// Required method for Designer support - do not modify
```

```
    /// the contents of this method with the code editor.
```

```
    /// </summary>
```

```
    private void InitializeComponent()
```

```
    {
```

```
        this.groupBox1 = new System.Windows.Forms.GroupBox();
```

```
        this.chkTurnInterconnectOnOff = new System.Windows.Forms.CheckBox();
```

```
        this.groupBox2 = new System.Windows.Forms.GroupBox();
```

```
        this.btnInsertComponent = new System.Windows.Forms.Button();
```

```
        this.btnInsertFeat = new System.Windows.Forms.Button();
```

```
        this.groupBox3 = new System.Windows.Forms.GroupBox();
```

```
        this.btnLoadFile = new System.Windows.Forms.Button();
```

```
        this.groupBox4 = new System.Windows.Forms.GroupBox();
```

```
        this.groupBox8 = new System.Windows.Forms.GroupBox();
```

```
        this.chkSolidBodies = new System.Windows.Forms.CheckBox();
```

```
        this.chkSurfaceBodies = new System.Windows.Forms.CheckBox();
```

```

this.chkPlanes = new System.Windows.Forms.CheckBox();
this.chkAxes = new System.Windows.Forms.CheckBox();
this.chkSketchedAndCurves = new System.Windows.Forms.CheckBox();
this.btnSetParams = new System.Windows.Forms.Button();
this.chkMaterial = new System.Windows.Forms.CheckBox();
this.chkCustomProps = new System.Windows.Forms.CheckBox();
this.chkIgnoreHiddenBodies = new System.Windows.Forms.CheckBox();
this.btn_IsInterFeat = new System.Windows.Forms.Button();
this.btnUpdate = new System.Windows.Forms.Button();
this.btnAuthorInfo = new System.Windows.Forms.Button();
this.btnIsUpdateNeeded = new System.Windows.Forms.Button();
this.btnGetParams = new System.Windows.Forms.Button();
this.btnGetRefFileName = new System.Windows.Forms.Button();
this.groupBox5 = new System.Windows.Forms.GroupBox();
this.btnClear = new System.Windows.Forms.Button();
this.richTextBox1 = new System.Windows.Forms.RichTextBox();
this.groupBox6 = new System.Windows.Forms.GroupBox();
this.btnBreakLink = new System.Windows.Forms.Button();
this.groupBox7 = new System.Windows.Forms.GroupBox();
this.btnModelExtRefs = new System.Windows.Forms.Button();
this.btnComponentExtRef = new System.Windows.Forms.Button();
this.btnFeatureExtRefs = new System.Windows.Forms.Button();
this.btnFormOpenFile = new System.Windows.Forms.Button();
this.btnCloseFile = new System.Windows.Forms.Button();
this.btnCloseApp = new System.Windows.Forms.Button();
this.groupBox1.SuspendLayout();
this.groupBox2.SuspendLayout();
this.groupBox3.SuspendLayout();
this.groupBox4.SuspendLayout();
this.groupBox8.SuspendLayout();
this.groupBox5.SuspendLayout();
this.groupBox6.SuspendLayout();
this.groupBox7.SuspendLayout();
this.SuspendLayout();
//
// groupBox1
//
this.groupBox1.Controls.Add(this.chkTurnInterconnectOnOff);
this.groupBox1.Location = new System.Drawing.Point(12, 12);
this.groupBox1.Name = "groupBox1";
this.groupBox1.Size = new System.Drawing.Size(148, 51);
this.groupBox1.TabIndex = 0;
this.groupBox1.TabStop = false;
this.groupBox1.Text = "On/Off";
//

```

```

        // chkTurnInterconnectOnOff
        //
        this.chkTurnInterconnectOnOff.AutoSize = true;
        this.chkTurnInterconnectOnOff.Location = new System.Drawing.Point(7, 20);
        this.chkTurnInterconnectOnOff.Name = "chkTurnInterconnectOnOff";
        this.chkTurnInterconnectOnOff.Size = new System.Drawing.Size(103, 17);
        this.chkTurnInterconnectOnOff.TabIndex = 0;
        this.chkTurnInterconnectOnOff.Text = "3D Interconnect";
        this.chkTurnInterconnectOnOff.UseVisualStyleBackColor = true;
        this.chkTurnInterconnectOnOff.CheckStateChanged += new
System.EventHandler(this.chkTurnInterconnectOnOff_CheckStateChanged);
        //
        // groupBox2
        //
        this.groupBox2.Controls.Add(this.btnInsertComponent);
        this.groupBox2.Controls.Add(this.btnInsertFeat);
        this.groupBox2.Location = new System.Drawing.Point(12, 166);
        this.groupBox2.Name = "groupBox2";
        this.groupBox2.Size = new System.Drawing.Size(148, 127);
        this.groupBox2.TabIndex = 1;
        this.groupBox2.TabStop = false;
        this.groupBox2.Text = "Inserting Third-Party Files";
        //
        // btnInsertComponent
        //
        this.btnInsertComponent.Location = new System.Drawing.Point(7, 70);
        this.btnInsertComponent.Name = "btnInsertComponent";
        this.btnInsertComponent.Size = new System.Drawing.Size(126, 34);
        this.btnInsertComponent.TabIndex = 1;
        this.btnInsertComponent.Text = "Insert as Component";
        this.btnInsertComponent.UseVisualStyleBackColor = true;
        this.btnInsertComponent.Click += new
System.EventHandler(this.btnInsertComponent_Click);
        //
        // btnInsertFeat
        //
        this.btnInsertFeat.Location = new System.Drawing.Point(6, 32);
        this.btnInsertFeat.Name = "btnInsertFeat";
        this.btnInsertFeat.Size = new System.Drawing.Size(127, 32);
        this.btnInsertFeat.TabIndex = 0;
        this.btnInsertFeat.Text = "Insert as Feature";
        this.btnInsertFeat.UseVisualStyleBackColor = true;
        this.btnInsertFeat.Click += new System.EventHandler(this.btnInsertFeat_Click);
        //
        // groupBox3
        //

```

```

        this.groupBox3.Controls.Add(this.btnLoadFile);
        this.groupBox3.Location = new System.Drawing.Point(12, 328);
        this.groupBox3.Name = "groupBox3";
        this.groupBox3.Size = new System.Drawing.Size(148, 81);
        this.groupBox3.TabIndex = 2;
        this.groupBox3.TabStop = false;
        this.groupBox3.Text = "Opening Third-Party Files";
        //
        // btnLoadFile
        //
        this.btnLoadFile.Location = new System.Drawing.Point(7, 31);
        this.btnLoadFile.Name = "btnLoadFile";
        this.btnLoadFile.Size = new System.Drawing.Size(126, 23);
        this.btnLoadFile.TabIndex = 0;
        this.btnLoadFile.Text = "Load File";
        this.btnLoadFile.UseVisualStyleBackColor = true;
        this.btnLoadFile.Click += new System.EventHandler(this.btnLoadFile_Click);
        //
        // groupBox4
        //
        this.groupBox4.Controls.Add(this.groupBox8);
        this.groupBox4.Controls.Add(this.btn_IsInterFeat);
        this.groupBox4.Controls.Add(this.btnUpdate);
        this.groupBox4.Controls.Add(this.btnAuthorInfo);
        this.groupBox4.Controls.Add(this.btnIsUpdateNeeded);
        this.groupBox4.Controls.Add(this.btnGetParams);
        this.groupBox4.Controls.Add(this.btnGetRefFileName);
        this.groupBox4.Location = new System.Drawing.Point(188, 12);
        this.groupBox4.Name = "groupBox4";
        this.groupBox4.Size = new System.Drawing.Size(296, 386);
        this.groupBox4.TabIndex = 3;
        this.groupBox4.TabStop = false;
        this.groupBox4.Text = "Editing Third-Party File Data";
    //
    // groupBox8
    //
    this.groupBox8.Controls.Add(this.chkSolidBodies);
    this.groupBox8.Controls.Add(this.chkSurfaceBodies);
    this.groupBox8.Controls.Add(this.chkPlanes);
    this.groupBox8.Controls.Add(this.chkAxes);
    this.groupBox8.Controls.Add(this.chkSketchedAndCurves);
    this.groupBox8.Controls.Add(this.btnSetParams);
    this.groupBox8.Controls.Add(this.chkMaterial);
    this.groupBox8.Controls.Add(this.chkCustomProps);
    this.groupBox8.Controls.Add(this.chkIgnoreHiddenBodies);

```

```
this.groupBox8.Location = new System.Drawing.Point(6, 114);
this.groupBox8.Name = "groupBox8";
this.groupBox8.Size = new System.Drawing.Size(283, 203);
this.groupBox8.TabIndex = 16;
this.groupBox8.TabStop = false;
this.groupBox8.Text = "Import Parameters";
//
// chkSolidBodies
//
this.chkSolidBodies.AutoSize = true;
this.chkSolidBodies.Location = new System.Drawing.Point(6, 30);
this.chkSolidBodies.Name = "chkSolidBodies";
this.chkSolidBodies.Size = new System.Drawing.Size(84, 17);
this.chkSolidBodies.TabIndex = 0;
this.chkSolidBodies.Text = "Solid Bodies";
this.chkSolidBodies.UseVisualStyleBackColor = true;
//
// chkSurfaceBodies
//
this.chkSurfaceBodies.AutoSize = true;
this.chkSurfaceBodies.Location = new System.Drawing.Point(141, 30);
this.chkSurfaceBodies.Name = "chkSurfaceBodies";
this.chkSurfaceBodies.Size = new System.Drawing.Size(98, 17);
this.chkSurfaceBodies.TabIndex = 1;
this.chkSurfaceBodies.Text = "Surface Bodies";
this.chkSurfaceBodies.UseVisualStyleBackColor = true;
//
// chkPlanes
//
this.chkPlanes.AutoSize = true;
this.chkPlanes.Location = new System.Drawing.Point(6, 63);
this.chkPlanes.Name = "chkPlanes";
this.chkPlanes.Size = new System.Drawing.Size(58, 17);
this.chkPlanes.TabIndex = 2;
this.chkPlanes.Text = "Planes";
this.chkPlanes.UseVisualStyleBackColor = true;
//
// chkAxes
//
this.chkAxes.AutoSize = true;
this.chkAxes.Location = new System.Drawing.Point(141, 63);
this.chkAxes.Name = "chkAxes";
this.chkAxes.Size = new System.Drawing.Size(49, 17);
this.chkAxes.TabIndex = 3;
this.chkAxes.Text = "Axes";
```

```

        this.chkAxes.UseVisualStyleBackColor = true;
        //
        // chkSketchedAndCurves
        //
        this.chkSketchedAndCurves.AutoSize = true;
        this.chkSketchedAndCurves.Location = new System.Drawing.Point(6, 95);
        this.chkSketchedAndCurves.Name = "chkSketchedAndCurves";
        this.chkSketchedAndCurves.Size = new System.Drawing.Size(129, 17);
        this.chkSketchedAndCurves.TabIndex = 4;
        this.chkSketchedAndCurves.Text = "Sketches And Curves";
        this.chkSketchedAndCurves.UseVisualStyleBackColor = true;
        //
        // btnSetParams
        //
        this.btnSetParams.Location = new System.Drawing.Point(73, 154);
        this.btnSetParams.Name = "btnSetParams";
        this.btnSetParams.Size = new System.Drawing.Size(117, 34);
        this.btnSetParams.TabIndex = 12;
        this.btnSetParams.Text = "Set Parameters";
        this.btnSetParams.UseVisualStyleBackColor = true;
        this.btnSetParams.Click += new System.EventHandler(this.btnSetParams_Click);
        //
        // chkMaterial
        //
        this.chkMaterial.AutoSize = true;
        this.chkMaterial.Location = new System.Drawing.Point(141, 95);
        this.chkMaterial.Name = "chkMaterial";
        this.chkMaterial.Size = new System.Drawing.Size(63, 17);
        this.chkMaterial.TabIndex = 6;
        this.chkMaterial.Text = "Material";
        this.chkMaterial.UseVisualStyleBackColor = true;
    //

        // chkCustomProps
        //
        this.chkCustomProps.AutoSize = true;
        this.chkCustomProps.Location = new System.Drawing.Point(6, 131);
        this.chkCustomProps.Name = "chkCustomProps";
        this.chkCustomProps.Size = new System.Drawing.Size(111, 17);
        this.chkCustomProps.TabIndex = 5;
        this.chkCustomProps.Text = "Custom Properties";
        this.chkCustomProps.UseVisualStyleBackColor = true;
        //
        // chkIgnoreHiddenBodies
        //
        this.chkIgnoreHiddenBodies.AutoSize = true;

```

```

this.chkIgnoreHiddenBodies.Location = new System.Drawing.Point(141, 131);
this.chkIgnoreHiddenBodies.Name = "chkIgnoreHiddenBodies";
this.chkIgnoreHiddenBodies.Size = new System.Drawing.Size(128, 17);
this.chkIgnoreHiddenBodies.TabIndex = 7;
this.chkIgnoreHiddenBodies.Text = "Ignore Hidden Bodies";
this.chkIgnoreHiddenBodies.UseVisualStyleBackColor = true;
//
// btn_IsInterFeat
//
this.btn_IsInterFeat.Location = new System.Drawing.Point(6, 19);
this.btn_IsInterFeat.Name = "btn_IsInterFeat";
this.btn_IsInterFeat.Size = new System.Drawing.Size(111, 34);
this.btn_IsInterFeat.TabIndex = 15;
this.btn_IsInterFeat.Text = "Is 3D Interconnect Feature?";
this.btn_IsInterFeat.UseVisualStyleBackColor = true;
this.btn_IsInterFeat.Click += new System.EventHandler(this.btn_IsInterFeat_Click);
//
// btnUpdate
//
this.btnUpdate.Location = new System.Drawing.Point(147, 335);
this.btnUpdate.Name = "btnUpdate";
this.btnUpdate.Size = new System.Drawing.Size(117, 34);
this.btnUpdate.TabIndex = 14;
this.btnUpdate.Text = "Update";
this.btnUpdate.UseVisualStyleBackColor = true;
this.btnUpdate.Click += new System.EventHandler(this.btnUpdate_Click);
//
// btnAuthorInfo
//
this.btnAuthorInfo.Location = new System.Drawing.Point(138, 70);
this.btnAuthorInfo.Name = "btnAuthorInfo";
this.btnAuthorInfo.Size = new System.Drawing.Size(132, 31);
this.btnAuthorInfo.TabIndex = 10;
this.btnAuthorInfo.Text = "Show Authoring Info";
this.btnAuthorInfo.UseVisualStyleBackColor = true;
this.btnAuthorInfo.Click += new System.EventHandler(this.btnAuthorInfo_Click);
//
// btnIsUpdateNeeded
//
this.btnIsUpdateNeeded.Location = new System.Drawing.Point(7, 335);
this.btnIsUpdateNeeded.Name = "btnIsUpdateNeeded";
this.btnIsUpdateNeeded.Size = new System.Drawing.Size(130, 34);
this.btnIsUpdateNeeded.TabIndex = 13;
this.btnIsUpdateNeeded.Text = "Needs Update?";
this.btnIsUpdateNeeded.UseVisualStyleBackColor = true;

```

```

        this.btnIsUpdateNeeded.Click += new
System.EventHandler(this.btnIsUpdateNeeded_Click);
//
// btnGetParams
//
this.btnGetParams.Location = new System.Drawing.Point(8, 68);
this.btnGetParams.Name = "btnGetParams";
this.btnGetParams.Size = new System.Drawing.Size(111, 34);
this.btnGetParams.TabIndex = 11;
this.btnGetParams.Text = "Show Parameters";
this.btnGetParams.UseVisualStyleBackColor = true;
this.btnGetParams.Click += new System.EventHandler(this.btnGetParams_Click);
//
// btnGetRefFileName
//
this.btnGetRefFileName.Location = new System.Drawing.Point(138, 20);
this.btnGetRefFileName.Name = "btnGetRefFileName";
this.btnGetRefFileName.Size = new System.Drawing.Size(132, 33);
this.btnGetRefFileName.TabIndex = 8;
this.btnGetRefFileName.Text = "Show Reference File";
this.btnGetRefFileName.UseVisualStyleBackColor = true;
this.btnGetRefFileName.Click += new System.EventHandler(this.btnGetRefFileName_Click);
//
// groupBox5
//
this.groupBox5.Controls.Add(this.btnClear);
this.groupBox5.Controls.Add(this.richTextBox1);
this.groupBox5.Location = new System.Drawing.Point(499, 12);
this.groupBox5.Name = "groupBox5";
this.groupBox5.Size = new System.Drawing.Size(325, 548);
this.groupBox5.TabIndex = 4;
this.groupBox5.TabStop = false;
this.groupBox5.Text = "Information";
//
// btnClear
//
this.btnClear.Location = new System.Drawing.Point(120, 473);
this.btnClear.Name = "btnClear";
this.btnClear.Size = new System.Drawing.Size(103, 33);
this.btnClear.TabIndex = 1;
this.btnClear.Text = "Clear";
this.btnClear.UseVisualStyleBackColor = true;
this.btnClear.Click += new System.EventHandler(this.btnClear_Click);
//
// richTextBox1
//

```

```

this.richTextBox1.BorderStyle = System.Windows.Forms.BorderStyle.None;
this.richTextBox1.Location = new System.Drawing.Point(6, 20);
this.richTextBox1.Name = "richTextBox1";
this.richTextBox1.Size = new System.Drawing.Size(313, 422);
this.richTextBox1.TabIndex = 0;
this.richTextBox1.Text = "";
//
// groupBox6
//
this.groupBox6.Controls.Add(this.btnBreakLink);
this.groupBox6.Location = new System.Drawing.Point(12, 430);
this.groupBox6.Name = "groupBox6";
this.groupBox6.Size = new System.Drawing.Size(148, 68);
this.groupBox6.TabIndex = 5;
this.groupBox6.TabStop = false;
this.groupBox6.Text = "Link";
//
// btnBreakLink
//
this.btnBreakLink.Location = new System.Drawing.Point(7, 30);
this.btnBreakLink.Name = "btnBreakLink";
this.btnBreakLink.Size = new System.Drawing.Size(126, 23);
this.btnBreakLink.TabIndex = 0;
this.btnBreakLink.Text = "Break Link";
this.btnBreakLink.UseVisualStyleBackColor = true;
this.btnBreakLink.Click += new System.EventHandler(this.btnBreakLink_Click);
//
// groupBox7
//
this.groupBox7.Controls.Add(this.btnModelExtRefs);
this.groupBox7.Controls.Add(this.btnComponentExtRef);
this.groupBox7.Controls.Add(this.btnFeatureExtRefs);
this.groupBox7.Location = new System.Drawing.Point(188, 432);
this.groupBox7.Name = "groupBox7";
this.groupBox7.Size = new System.Drawing.Size(283, 66);
this.groupBox7.TabIndex = 6;
this.groupBox7.TabStop = false;
this.groupBox7.Text = "External References";
//
// btnModelExtRefs
//
this.btnModelExtRefs.Location = new System.Drawing.Point(188, 26);
this.btnModelExtRefs.Name = "btnModelExtRefs";
this.btnModelExtRefs.Size = new System.Drawing.Size(85, 34);
this.btnModelExtRefs.TabIndex = 2;

```

```

this.btnModelExtRefs.Text = "Model";
this.btnModelExtRefs.UseVisualStyleBackColor = true;
this.btnModelExtRefs.Click += new System.EventHandler(this.btnModelExtRefs_Click);
//
// btnComponentExtRef
//
this.btnComponentExtRef.Location = new System.Drawing.Point(97, 26);
this.btnComponentExtRef.Name = "btnComponentExtRef";
this.btnComponentExtRef.Size = new System.Drawing.Size(85, 34);
this.btnComponentExtRef.TabIndex = 1;
this.btnComponentExtRef.Text = "Component";
this.btnComponentExtRef.UseVisualStyleBackColor = true;
this.btnComponentExtRef.Click += new
System.EventHandler(this.btnComponentExtRef_Click);
//
// btnFeatureExtRefs
//
this.btnFeatureExtRefs.Location = new System.Drawing.Point(7, 26);
this.btnFeatureExtRefs.Name = "btnFeatureExtRefs";
this.btnFeatureExtRefs.Size = new System.Drawing.Size(85, 34);
this.btnFeatureExtRefs.TabIndex = 0;
this.btnFeatureExtRefs.Text = "Feature";
this.btnFeatureExtRefs.UseVisualStyleBackColor = true;
this.btnFeatureExtRefs.Click += new
System.EventHandler(this.btnFeatureExtRefs_Click);
//
// btnFormOpenFile
//
this.btnFormOpenFile.Location = new System.Drawing.Point(12, 92);
this.btnFormOpenFile.Name = "btnFormOpenFile";
this.btnFormOpenFile.Size = new System.Drawing.Size(126, 39);
this.btnFormOpenFile.TabIndex = 7;
this.btnFormOpenFile.Text = "Open SOLIDWORKS File";
this.btnFormOpenFile.UseVisualStyleBackColor = true;
this.btnFormOpenFile.Click += new System.EventHandler(this.btnFormOpenFile_Click);
//
// btnCloseFile
//
this.btnCloseFile.Location = new System.Drawing.Point(92, 521);
this.btnCloseFile.Name = "btnCloseFile";
this.btnCloseFile.Size = new System.Drawing.Size(103, 39);
this.btnCloseFile.TabIndex = 8;
this.btnCloseFile.Text = "Close File";
this.btnCloseFile.UseVisualStyleBackColor = true;
this.btnCloseFile.Click += new System.EventHandler(this.btnCloseFile_Click);
//

```

```

        // btnCloseApp
        //
        this.btnCloseApp.Location = new System.Drawing.Point(236, 521);
        this.btnCloseApp.Name = "btnCloseApp";
        this.btnCloseApp.Size = new System.Drawing.Size(103, 39);
        this.btnCloseApp.TabIndex = 9;
        this.btnCloseApp.Text = "Close Application";
        this.btnCloseApp.UseVisualStyleBackColor = true;
        this.btnCloseApp.Click += new System.EventHandler(this.btnCloseApp_Click);
        //
        // InterConnectingApp
        //
        this.AutoScaleDimensions = new System.Drawing.SizeF(6F, 13F);
        this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
        this.ClientSize = new System.Drawing.Size(842, 579);
        this.Controls.Add(this.btnCloseApp);
        this.Controls.Add(this.btnCloseFile);
        this.Controls.Add(this.btnFormOpenFile);
        this.Controls.Add(this.groupBox7);
        this.Controls.Add(this.groupBox6);
        this.Controls.Add(this.groupBox5);
        this.Controls.Add(this.groupBox4);
        this.Controls.Add(this.groupBox3);
        this.Controls.Add(this.groupBox2);
        this.Controls.Add(this.groupBox1);
        this.FormBorderStyle = System.Windows.Forms.FormBorderStyle.FixedSingle;
        this.Name = "InterConnectingApp";
        this.Text = "InterConnectingApp";
        this.groupBox1.ResumeLayout(false);
        this.groupBox1.PerformLayout();
        this.groupBox2.ResumeLayout(false);
        this.groupBox3.ResumeLayout(false);
        this.groupBox4.ResumeLayout(false);
        this.groupBox8.ResumeLayout(false);
        this.groupBox8.PerformLayout();
        this.groupBox5.ResumeLayout(false);
        this.groupBox6.ResumeLayout(false);
        this.groupBox7.ResumeLayout(false);
        this.ResumeLayout(false);
    }

#endregion

private System.Windows.Forms.GroupBox groupBox1;
private System.Windows.Forms.CheckBox chkTurnInterconnectOnOff;

```

```

private System.Windows.Forms.GroupBox groupBox2;
private System.Windows.Forms.Button btnInsertComponent;
private System.Windows.Forms.Button btnInsertFeat;
private System.Windows.Forms.GroupBox groupBox3;
private System.Windows.Forms.Button btnLoadFile;
private System.Windows.Forms.GroupBox groupBox4;
private System.Windows.Forms.Button btnAuthorInfo;
private System.Windows.Forms.Button btnGetRefFileName;
private System.Windows.Forms.CheckBox chkIgnoreHiddenBodies;
private System.Windows.Forms.CheckBox chkMaterial;
private System.Windows.Forms.CheckBox chkCustomProps;
private System.Windows.Forms.CheckBox chkSketchedAndCurves;
private System.Windows.Forms.CheckBox chkAxes;
private System.Windows.Forms.CheckBox chkPlanes;
private System.Windows.Forms.CheckBox chkSurfaceBodies;
private System.Windows.Forms.CheckBox chkSolidBodies;
private System.Windows.Forms.Button btnUpdate;
private System.Windows.Forms.Button btnIsUpdateNeeded;
private System.Windows.Forms.Button btnSetParams;
private System.Windows.Forms.Button btnGetParams;
private System.Windows.Forms.GroupBox groupBox5;
private System.Windows.Forms.RichTextBox richTextBox1;
private System.Windows.Forms.Button btnClear;
private System.Windows.Forms.GroupBox groupBox6;
private System.Windows.Forms.Button btnBreakLink;
private System.Windows.Forms.GroupBox groupBox7;
private System.Windows.Forms.Button btnModelExtRefs;
private System.Windows.Forms.Button btnComponentExtRef;
private System.Windows.Forms.Button btnFeatureExtRefs;
private System.Windows.Forms.Button btnFormOpenFile;
private System.Windows.Forms.Button btnCloseFile;
private System.Windows.Forms.Button btnCloseApp;
private System.Windows.Forms.Button btn_IsInterFeat;
private System.Windows.Forms.GroupBox groupBox8;
}
}

```

//InterConnectingApp.cs:

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;

```

```

using System.Threading.Tasks;
using System.Windows.Forms;
using System.IO;
using SolidWorks.Interop.sldworks;
using SolidWorks.Interop.swconst;

namespace Interconnect
{
    public partial class InterConnectingApp : Form
    {
        public SldWorks m_SwApp = default(SldWorks);
        public ModelDoc2 swModelDoc = default(ModelDoc2);
        public PartDoc swPart = default(PartDoc);
        public AssemblyDoc swAssembly = default(AssemblyDoc);
        public Feature swFeat = default(Feature);
        public Component2 swComp = default(Component2);
        public Import3DInterconnectData swIntrConData = default(Import3DInterconnectData);
        public SelectionMgr swSelMgr = default(SelectionMgr);
        public string strCurModelDoc = string.Empty;
        public int m_LongStatus;
        public int m_Warning;
        public bool m_BooleanStatus;
        public StringBuilder strEventData;

        delegate void StringArgReturningVoidDelegate(string text);

        public InterConnectingApp()
        {
            InitializeComponent();
        }

        public void RunInterconnectApplication(SldWorks swApp)
        {
            m_SwApp = swApp;
            swModelDoc = (ModelDoc2)m_SwApp.ActiveDoc;

            if (swModelDoc != null)
            {
                strCurModelDoc = swModelDoc.GetPathName();
                string strExt = Path.GetExtension(strCurModelDoc);

                if (strExt.ToLower() == ".sldprt")
                    swPart = (PartDoc)swModelDoc;
                else if (strExt.ToLower() == ".sldasm")

```

```

        swAssembly = (AssemblyDoc)swModelDoc;
    }

    AttachEventHandlers();
    strEventData = new StringBuilder();

    Application.Run(this);
}

public void AttachEventHandlers()
{
    AttachSWEvents();
}

public void AttachSWEvents()
{
    m_SwApp.Begin3DInterconnectTranslationNotify +=
M_SwApp_Begin3DInterconnectTranslationNotify;
    m_SwApp.End3DInterconnectTranslationNotify +=
M_SwApp_End3DInterconnectTranslationNotify;
}

public int M_SwApp_End3DInterconnectTranslationNotify(string FileName)
{
    strEventData = strEventData.Append("3D Interconnect translation finished");
    strEventData = strEventData.Append("\r\n");

    return 0;
}

public int M_SwApp_Begin3DInterconnectTranslationNotify(string FileName)
{
    strEventData = strEventData.Append("3D Interconnect translation started");
    strEventData = strEventData.Append("\r\n");

    return 0;
}

public int BeginTranslationNotify()
{
    int i = 0;
    return i;
}

//Information field
public void SetRichTextBoxVal(string strText)
{
    try

```

```

    {
        if (this.richTextBox1.InvokeRequired == true)
        {
            StringArgReturningVoidDelegate d = new
StringArgReturningVoidDelegate(SetRichTextBoxVal);
            this.Invoke(d, new object[] { strText });
        }
        else
        {
            this.richTextBox1.Text = strText;
        }
    }
    catch (Exception e)
    {
        MessageBox.Show(e.Message);
    }
}

//Turn 3D Interconnect on and off
private void chkTurnInterconnectOnOff_CheckStateChanged(object sender, EventArgs e)
{
    if (chkTurnInterconnectOnOff.CheckState == CheckState.Checked)
    {
        m_SwApp.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swMultiCAD_Enable3DInterconnect,
true);
    }
    else if (chkTurnInterconnectOnOff.CheckState == CheckState.Unchecked)
    {
        m_SwApp.SetUserPreferenceToggle((int)swUserPreferenceToggle_e.swMultiCAD_Enable3DInterconnect,
false);
    }
}

//Open a SOLIDWORKS Design document
private void btnFormOpenFile_Click(object sender, EventArgs e)
{
    OpenFileDialog m_fileOpenDlg = new OpenFileDialog();
    m_fileOpenDlg.Filter = "SolidWorks File | *.sldprt; *.sldasm";
    m_fileOpenDlg.Title = "Open SolidWorks Design File";

    string FilePath = string.Empty;
    string FileExtension = string.Empty;

    if (m_fileOpenDlg.ShowDialog() == DialogResult.OK)
    {

```

```

        FilePath = m_fileOpenDlg.FileName;
        FileExtension = Path.GetExtension(FilePath);

        if (FileExtension.ToLower() == ".sldasm")
        {
            swModelDoc = (ModelDoc2)m_SwApp.OpenDoc6(FilePath,
            (int)swDocumentTypes_e.swDocASSEMBLY, (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref
            m_LongStatus, ref m_Warning);
            swAssembly = (AssemblyDoc)swModelDoc;
        }
        else if (FileExtension.ToLower() == ".sldprt")
        {
            swModelDoc = (ModelDoc2)m_SwApp.OpenDoc6(FilePath,
            (int)swDocumentTypes_e.swDocPART, (int)swOpenDocOptions_e.swOpenDocOptions_Silent, "", ref
            m_LongStatus, ref m_Warning);
            swPart = (PartDoc)swModelDoc;
        }
    }
}

//Insert third-party feature into the current part
private void btnInsertFeat_Click(object sender, EventArgs e)
{
    swModelDoc = (ModelDoc2)m_SwApp.ActiveDoc;
    if (swModelDoc != null) {
        if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocPART)
        {
            if (swPart != null)
            {
                OpenFileDialog m_fileOpenDlg = new OpenFileDialog();
                //m_fileOpenDlg.Filter = "Parasolid (*.x_t; *.x_b; *.xmt_txt;
                *.xmt_bin) | IGES (*.igs; *.iges) | ACIS (*.sat) | STEP (*.step; *.stp) | VDAFS (*.vda) | VRML
                (*.wrl) | STL (*.stl) | CGR (*.cgr) | 3MF (*.3mf) | Rhino (*.3dm)";
                m_fileOpenDlg.Title = "Insert Third-Party Feature";

                string FilePath = string.Empty;

                if (m_fileOpenDlg.ShowDialog() == DialogResult.OK)
                {
                    FilePath = m_fileOpenDlg.FileName;
                    swFeat = (Feature)swPart.InsertImportedFeature(FilePath, out
m_LongStatus);

                    if (swFeat == null)
                        SetRichTextBoxVal(m_LongStatus.ToString());
                }
            }
        }
    }
}

```

```

    }

    //Insert third-party component into the current assembly
    private void btnInsertComponent_Click(object sender, EventArgs e)
    {
        swModelDoc = (ModelDoc2)m_SwApp.ActiveDoc;
        if (swModelDoc != null)
        {
            if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocASSEMBLY)
            {
                if (swAssembly != null)
                {
                    OpenFileDialog m_fileOpenDlg = new OpenFileDialog();
                    m_fileOpenDlg.Filter = "All Files | *.*";
                    m_fileOpenDlg.Title = "Insert Third-Party Component";

                    string FilePath = string.Empty;

                    if (m_fileOpenDlg.ShowDialog() == DialogResult.OK)
                    {
                        FilePath = m_fileOpenDlg.FileName;
                        Object InsertedComp = null;
                        int retVal = (int)swAssembly.InsertImportedComponent(FilePath, 0,
0, 0, out InsertedComp);

                        swComp = (InsertedComp != null) ? (Component2)InsertedComp : null;
                        if (swComp == null)
                            SetRichTextBoxVal(retVal.ToString());
                        else
                        {
                            SetRichTextBoxVal(strEventData.ToString());
                            strEventData.Length = 0;
                        }
                    }
                }
            }
        }
    }

    //Open a third-party file into a new document
    private void btnLoadFile_Click(object sender, EventArgs e)
    {
        OpenFileDialog m_fileOpenDlg = new OpenFileDialog();
        m_fileOpenDlg.Filter = "All Files | *.*";
        m_fileOpenDlg.Title = "Open Third-Party File";

        string FilePath = string.Empty;

```

```

        if (m_fileOpenDlg.ShowDialog() == DialogResult.OK)
        {
            FilePath = m_fileOpenDlg.FileName;

            swModelDoc = m_SwApp.LoadFile4(FilePath, "r", null, ref m_LongStatus);

            string strLoadFileError = "";
            if (m_LongStatus != 0)
                strLoadFileError = "File load error as defined in swFileLoadError_e: " +
m_LongStatus.ToString();

            StringBuilder strLoadFileInfo = new StringBuilder();
            strLoadFileInfo = strLoadFileInfo.Append(strLoadFileError);

            strLoadFileInfo = strLoadFileInfo.Append("\r\n");
            strLoadFileInfo = strLoadFileInfo.Append(strEventData.ToString());
            SetRichTextBoxVal(strLoadFileInfo.ToString());
            strEventData.Length = 0;

            if (swModelDoc != null)
            {
                string strModelName = swModelDoc.GetPathName();
                string strExt = Path.GetExtension(strModelName);

                if (strExt.ToLower() == ".sldprt")
                    swPart = (PartDoc)swModelDoc;
                else if (strExt.ToLower() == ".sldasm")
                    swAssembly = (AssemblyDoc)swModelDoc;
            }
        }
    }

    private void btnCloseApp_Click(object sender, EventArgs e)
    {
        Application.Exit();
    }

    private void btnCloseFile_Click(object sender, EventArgs e)
    {
        m_SwApp.CloseDoc(strCurModelDoc);
    }

    //Get whether the selected feature is a 3D Interconnect feature
    private void btn_IsInterFeat_Click(object sender, EventArgs e)
    {
        if (swModelDoc != null)
        {

```

```

swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;

m_BoolStatus = false;

if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocPART)
{
    swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
    if (swFeat != null)
        m_BoolStatus = swFeat.Is3DInterconnectFeature;
    else
        MessageBox.Show("No feature selected");
}
else if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocASSEMBLY)
{
    object objSelected = swSelMgr.GetSelectedObject6(1, -1);
    string strSelectedObjName;
    string sType;
    int iType;
    swSelMgr.GetSelectByIdSpecification(objSelected, out strSelectedObjName,
    out sType, out iType);
    swAssembly = (AssemblyDoc)swModelDoc;
    if (strSelectedObjName.Length > 0)
    {
        string[] vNames = strSelectedObjName.Split('@');
        swFeat = (Feature)swAssembly.FeatureByName(vNames[0]);

        if (swFeat == null && (iType != 22 || iType != 76))
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);

        if (swFeat != null)
            m_BoolStatus = swFeat.Is3DInterconnectFeature;
        else
            MessageBox.Show("No feature selected");
    }
}

if (m_BoolStatus == false)
    SetRichTextBoxVal("This is NOT a 3D Interconnect feature");
else
    SetRichTextBoxVal("This is a 3D Interconnect feature");
}

private void btnClear_Click(object sender, EventArgs e)
{
    SetRichTextBoxVal("");
}

```

```

}

//Show the selected item's reference file name
private void btnGetRefFileName_Click(object sender, EventArgs e)
{
    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;
        swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        if ((swFeat != null) & (swFeat.Is3DInterconnectFeature))
        {
            object DataObj = null;
            swFeat.GetImportedFeatureParameters(out DataObj);
            swIntrConData = (Import3DInterconnectData)DataObj;

            if (swIntrConData != null)
            {
                SetRichTextBoxVal(swIntrConData.GetReferencedFileName());
            }
        }
    }
}

//Show the selected item's authoring application
private void btnAuthorInfo_Click(object sender, EventArgs e)
{
    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;
        swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        if ((swFeat != null) & (swFeat.Is3DInterconnectFeature))
        {
            object DataObj = null;
            swFeat.GetImportedFeatureParameters(out DataObj);
            swIntrConData = (Import3DInterconnectData)DataObj;

            if (swIntrConData != null)
            {
                SetRichTextBoxVal(swIntrConData.GetAuthoringInfo());
            }
        }
    }
}

//Show the import settings

```

```

private void btnGetParams_Click(object sender, EventArgs e)
{
    StringBuilder strBld = new StringBuilder();
    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;

        if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocPART)
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        else if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocASSEMBLY)
        {
            object objSelected = swSelMgr.GetSelectedObject6(1, -1);
            string strSelectedObjName, sType;
            int iType;
            swSelMgr.GetSelectByIdSpecification(objSelected, out strSelectedObjName,
            out sType, out iType);
            swAssembly = (AssemblyDoc)swModelDoc;
            if (strSelectedObjName.Length > 0)
            {
                string[] vNames = strSelectedObjName.Split('@');
                swFeat = (Feature)swAssembly.FeatureByName(vNames[0]);

                if (swFeat == null && (iType != 22 || iType != 76))
                    swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            }
        }

        if ((swFeat != null) & (swFeat.Is3DInterconnectFeature))
        {
            object DataObj = null;
            swFeat.GetImportedFeatureParameters(out DataObj);
            swIntrConData = (Import3DInterconnectData)DataObj;

            if (swIntrConData != null)
            {
                if (swIntrConData.SolidBodies == true)
                {
                    strBld = strBld.Append("Solid Bodies"); strBld =
strBld.Append("\r\n");
                }
                if (swIntrConData.SurfaceBodies == true)
                {
                    strBld = strBld.Append("Surface Bodies"); strBld =
strBld.Append("\r\n");
                }
                if (swIntrConData.Planes == true)
                {

```

```

        strBld = strBld.Append("Planes"); strBld = strBld.Append("\r\n");
    }
    if (swIntrConData.Axes == true)
    {
        strBld = strBld.Append("Axes"); strBld = strBld.Append("\r\n");
    }
    if (swIntrConData.SketchesAndCurves == true)
    {
        strBld = strBld.Append("Sketches And Curves"); strBld =
strBld.Append("\r\n");
    }
    if (swIntrConData.CustomProperties == true)
    {
        strBld = strBld.Append("Custom Properties"); strBld =
strBld.Append("\r\n");
    }
    if (swIntrConData.Material == true)
    {
        strBld = strBld.Append("Material"); strBld =
strBld.Append("\r\n");
    }
    if (swIntrConData.IgnoreHiddenEntities == true)
    {
        strBld = strBld.Append("Ignore Hidden Entities"); strBld =
strBld.Append("\r\n");
    }

    SetRichTextBoxVal(strBld.ToString());
}
}
}

//Set new import settings
private void btnSetParams_Click(object sender, EventArgs e)
{
    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;

        if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocPART)
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        else if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocASSEMBLY)
        {
            object objSelected = swSelMgr.GetSelectedObject6(1, -1);
            string strSelectedObjName, sType;
            int iType;

```

```

        swSelMgr.GetSelectByIdSpecification(objSelected, out strSelectedObjName,
out sType, out iType);
        swAssembly = (AssemblyDoc)swModelDoc;
        if (strSelectedObjName.Length > 0)
        {
            string[] vNames = strSelectedObjName.Split('@');
            swFeat = (Feature)swAssembly.FeatureByName(vNames[0]);

            if (swFeat == null && (iType != 22 || iType != 76))
                swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        }
    }

    if ((swFeat != null) & (swFeat.Is3DInterconnectFeature))
    {
        object DataObj = null;
        swFeat.GetImportedFeatureParameters(out DataObj);
        swIntrConData = (Import3DInterconnectData)DataObj;

        if (swIntrConData != null)
        {
            if (chkSolidBodies.CheckState == CheckState.Checked)
                swIntrConData.SolidBodies = true;
            else
                swIntrConData.SolidBodies = false;

            if (chkSurfaceBodies.CheckState == CheckState.Checked)
                swIntrConData.SurfaceBodies = true;
            else
                swIntrConData.SurfaceBodies = false;

            if (chkPlanes.CheckState == CheckState.Checked)
                swIntrConData.Planes = true;
            else
                swIntrConData.Planes = false;

            if (chkAxes.CheckState == CheckState.Checked)
                swIntrConData.Axes = true;
            else
                swIntrConData.Axes = false;

            if (chkSketchedAndCurves.CheckState == CheckState.Checked)
                swIntrConData.SketchesAndCurves = true;
            else
                swIntrConData.SketchesAndCurves = false;

            if (chkCustomProps.CheckState == CheckState.Checked)

```

```

        swIntrConData.CustomProperties = true;
    else
        swIntrConData.CustomProperties = false;

    if (chkMaterial.CheckState == CheckState.Checked)
        swIntrConData.Material = true;
    else
        swIntrConData.Material = false;

    if (chkIgnoreHiddenBodies.CheckState == CheckState.Checked)
        swIntrConData.IgnoreHiddenEntities = true;
    else
        swIntrConData.IgnoreHiddenEntities = false;

    m_LongStatus = swFeat.SetImportedFeatureParameters(swIntrConData);

    if (m_LongStatus != 0)
    {
        SetRichTextBoxVal("Error setting import parameters as defined in
sw3DInterconnectImportErrors_e: " + m_LongStatus.ToString());
    }
}

}

}

//Get whether the selected 3D Interconnect item needs an update
private void btnIsUpdateNeeded_Click(object sender, EventArgs e)
{
    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;

        if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocPART)
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        else if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocASSEMBLY)
        {
            object objSelected = swSelMgr.GetSelectedObject6(1, -1);
            string strSelectedObjName, sType;
            int iType;
            swSelMgr.GetSelectByIdSpecification(objSelected, out strSelectedObjName,
out sType, out iType);
            swAssembly = (AssemblyDoc)swModelDoc;
            if (strSelectedObjName.Length > 0)
            {
                string[] vNames = strSelectedObjName.Split('@');

```

```

        swFeat = (Feature)swAssembly.FeatureByName(vNames[0]);

        if (swFeat == null && (iType != 22 || iType != 76))
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
    }
}

if ((swFeat != null) & (swFeat.Is3DInterconnectFeature))
{
    if (swFeat.Is3DInterconnectUpdateAvailable == true)
        SetRichTextBoxVal("Need to update");
    else
        SetRichTextBoxVal("No update needed");
}
}

//Update the selected 3D Interconnect item
private void btnUpdate_Click(object sender, EventArgs e)
{
    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;

        if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocPART)
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        else if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocASSEMBLY)
        {
            object objSelected = swSelMgr.GetSelectedObject6(1, -1);
            string strSelectedObjName, sType;
            int iType;
            swSelMgr.GetSelectByIdSpecification(objSelected, out strSelectedObjName,
            out sType, out iType);
            swAssembly = (AssemblyDoc)swModelDoc;
            if (strSelectedObjName.Length > 0)
            {
                string[] vNames = strSelectedObjName.Split('@');
                swFeat = (Feature)swAssembly.FeatureByName(vNames[0]);

                if (swFeat == null && (iType != 22 || iType != 76))
                    swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            }
        }

        if ((swFeat != null) & (swFeat.Is3DInterconnectFeature))
        {

```

```

        bool isUpdated = swFeat.Update3DInterconnectModel();
        if (isUpdated)
            SetRichTextBoxVal("Updated");
        else
            SetRichTextBoxVal("Error during update");
    }
}

//Break the link between the selected item and its original file
private void btnBreakLink_Click(object sender, EventArgs e)
{
    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;

        if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocPART)
            swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        else if (swModelDoc.GetType() == (int)swDocumentTypes_e.swDocASSEMBLY)
        {
            object objSelected = swSelMgr.GetSelectedObject6(1, -1);
            string strSelectedObjName, sType;
            int iType;
            swSelMgr.GetSelectByIdSpecification(objSelected, out strSelectedObjName,
            out sType, out iType);
            swAssembly = (AssemblyDoc)swModelDoc;
            if (strSelectedObjName.Length > 0)
            {
                string[] vNames = strSelectedObjName.Split('@');
                swFeat = (Feature)swAssembly.FeatureByName(vNames[0]);

                if (swFeat == null && (iType != 22 || iType != 76))
                    swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
            }
        }

        if ((swFeat != null) & (swFeat.Is3DInterconnectFeature))
        {
            bool Silent = false; bool allComp = false;

            DialogResult dRes = MessageBox.Show("Silent?", "Want Silent",
            MessageBoxButtons.YesNo);
            switch (dRes)
            {
                case DialogResult.Yes:
                {
                    Silent = true;

```

```

        break;
    }
    case DialogResult.No:
    {
        Silent = false;
        break;
    }
}

DialogResult dRes1 = MessageBox.Show("All Components?", "All Comp",
MessageBoxButtons.YesNo);
switch (dRes1)
{
    case DialogResult.Yes:
    {
        allComp = true;
        break;
    }
    case DialogResult.No:
    {
        allComp = false;
        break;
    }
}

int lError = swFeat.BreakLink(allComp, Silent);
string inp = "Break Link Error" + lError.ToString();
SetRichTextBoxVal(inp);
}
}
}

```

```

private void btnFeatureExtRefs_Click(object sender, EventArgs e)
{
    object vModelPathName = null;
    object vComponentPathName = null;
    object vFeature = null;
    object vDataType = null;
    object vStatus = null;
    object vRefEntity = null;
    object vFeatComp = null;
    int nConfigOpt = 0;
    string sConfigName = null;
    int nRefCount = 0;
    int nSelType = 0;
}

```

```

int i = 0;

if (swModelDoc != null)
{
    swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;
    nSelType = swSelMgr.GetSelectedObjectType3(1, -1);

    if (nSelType != (int)swSelectType_e.swSelCOMPONENTS && nSelType != 0)
    {
        swFeat = (Feature)swSelMgr.GetSelectedObject6(1, -1);
        nRefCount = swFeat.ListExternalFileReferencesCount();
        swFeat.ListExternalFileReferences2(out vModelPathName, out
vComponentPathName, out vFeature, out vDataType, out vStatus, out vRefEntity, out vFeatComp,
out nConfigOpt, out sConfigName);

        if (nRefCount >= 1)
        {
            object[] ModelPathName = new object[nRefCount - 1];
            object[] ComponentPathName = new object[nRefCount - 1];
            object[] Feature = new object[nRefCount - 1];
            object[] DataType = new object[nRefCount - 1];
            int[] Status = new int[nRefCount - 1];
            object[] RefEntity = new object[nRefCount - 1];
            object[] FeatComp = new object[nRefCount - 1];

            ModelPathName = (object[])vModelPathName;
            ComponentPathName = (object[])vComponentPathName;
            Feature = (object[])vFeature;
            DataType = (object[])vDataType;
            Status = (int[])vStatus;
            RefEntity = (object[])vRefEntity;
            FeatComp = (object[])vFeatComp;

            StringBuilder sb = new StringBuilder();

            for (i = 0; i <= nRefCount - 1; i++)
            {
                sb = sb.Append("=>"); sb = sb.Append("Reference Count = "); sb =
sb.Append(nRefCount.ToString()); sb = sb.Append("\r\n");
                sb = sb.Append(" "); sb = sb.Append("\r\n");
                sb = sb.Append("=>"); sb = sb.Append("Model path + name = "); sb =
sb.Append(ModelPathName[i].ToString()); sb = sb.Append("\r\n");
                sb = sb.Append("=>"); sb = sb.Append("Component path + name = ");
sb = sb.Append(ComponentPathName[i].ToString()); sb = sb.Append("\r\n");
                sb = sb.Append("=>"); sb = sb.Append("Feature = "); sb =
sb.Append(Feature[i].ToString()); sb = sb.Append("\r\n");
                sb = sb.Append("=>"); sb = sb.Append("Data type = "); sb =
sb.Append(DataType[i].ToString()); sb = sb.Append("\r\n");
            }
        }
    }
}

```

```

        sb = sb.Append("=>"); sb = sb.Append("Status = "); sb =
sb.Append(System.Convert.ToString(Status[i])); sb = sb.Append("\r\n");

        sb = sb.Append("=>"); sb = sb.Append("Reference entity = "); sb =
sb.Append(RefEntity[i].ToString()); sb = sb.Append("\r\n");

        sb = sb.Append("=>"); sb = sb.Append("Feature component = "); sb =
sb.Append(FeatComp[i].ToString()); sb = sb.Append("\r\n");

        sb = sb.Append("=>"); sb = sb.Append("Config option = "); sb =
sb.Append(nConfigOpt.ToString()); sb = sb.Append("\r\n");

        sb = sb.Append("=>"); sb = sb.Append("Config name = "); sb =
sb.Append(sConfigName); sb = sb.Append("\r\n");

        SetRichTextBoxVal(sb.ToString());

    }

}

}

}

private void btnComponentExtRef_Click(object sender, EventArgs e)
{
    object vModelPathName = null;
    object vComponentPathName = null;
    object vFeature = null;
    object vDataType = null;
    object vStatus = null;
    object vRefEntity = null;
    object vFeatComp = null;
    int nConfigOpt = 0;
    string sConfigName = null;
    int nRefCount = 0;
    int nSelType = 0;
    int i = 0;

    if (swModelDoc != null)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;
        nSelType = swSelMgr.GetSelectedObjectType3(1, -1);

        if (nSelType == (int)swSelectType_e.swSelCOMPONENTS)
        {
            swComp = (Component2)swSelMgr.GetSelectedObject6(1, -1);
            nRefCount = swComp.ListExternalFileReferencesCount();

            swComp.ListExternalFileReferences2(out vModelPathName, out
vComponentPathName, out vFeature, out vDataType, out vStatus, out vRefEntity, out vFeatComp,
out nConfigOpt, out sConfigName);

            if (nRefCount >= 1)
            {
                object[] ModelPathName = new object[nRefCount - 1];

```

```

        object[] ComponentPathName = new object[nRefCount - 1];
        object[] Feature = new object[nRefCount - 1];
        object[] DataType = new object[nRefCount - 1];
        int[] Status = new int[nRefCount - 1];
        object[] RefEntity = new object[nRefCount - 1];
        object[] FeatComp = new object[nRefCount - 1];

        ModelPathName = (object[])vModelPathName;
        ComponentPathName = (object[])vComponentPathName;
        Feature = (object[])vFeature;
        DataType = (object[])vDataType;
        Status = (int[])vStatus;
        RefEntity = (object[])vRefEntity;
        FeatComp = (object[])vFeatComp;

        StringBuilder sb = new StringBuilder();

        for (i = 0; i <= nRefCount - 1; i++)
        {
            sb = sb.Append("=>"); sb = sb.Append("Reference Count = "); sb =
sb.Append(nRefCount.ToString()); sb = sb.Append("\r\n");
            sb = sb.Append(" "); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Model path + name = "); sb =
sb.Append(ModelPathName[i].ToString()); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Component path + name = ");
sb = sb.Append(ComponentPathName[i].ToString()); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Feature = "); sb =
sb.Append(Feature[i].ToString()); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Data type = "); sb =
sb.Append(DataType[i].ToString()); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Status = "); sb =
sb.Append(System.Convert.ToString(Status[i])); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Reference entity = "); sb =
sb.Append(RefEntity[i].ToString()); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Feature component = "); sb =
sb.Append(FeatComp[i].ToString()); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Config option = "); sb =
sb.Append(nConfigOpt.ToString()); sb = sb.Append("\r\n");
            sb = sb.Append("=>"); sb = sb.Append("Config name = "); sb =
sb.Append(sConfigName); sb = sb.Append("\r\n");

            SetRichTextBoxVal(sb.ToString());
        }
    }
    else
        SetRichTextBoxVal("This is not a component");
}
}

```

```

private void btnModelExtRefs_Click(object sender, EventArgs e)
{
    object vModelPathName = null;
    object vComponentPathName = null;
    object vFeature = null;
    object vDataType = null;
    object vStatus = null;
    object vRefEntity = null;
    object vFeatComp = null;
    int nConfigOpt = 0;
    string sConfigName = null;
    int nRefCount = 0;
    int nSelType = 0;
    int i = 0;

    if (swModelDoc != null && swModelDoc.GetType() ==
(int)swDocumentTypes_e.swDocPART)
    {
        swSelMgr = (SelectionMgr)swModelDoc.SelectionManager;
        nSelType = swSelMgr.GetSelectedObjectType3(1, -1);

        if (nSelType != (int)swSelectType_e.swSelCOMPONENTS)
        {
            nRefCount = swModelDoc.Extension.ListExternalFileReferencesCount();
            swModelDoc.Extension.ListExternalFileReferences(out vModelPathName, out
vComponentPathName, out vFeature, out vDataType, out vStatus, out vRefEntity, out vFeatComp,
out nConfigOpt, out sConfigName);

            if (nRefCount >= 1)
            {
                object[] ModelPathName = new object[nRefCount - 1];
                object[] ComponentPathName = new object[nRefCount - 1];
                object[] Feature = new object[nRefCount - 1];
                object[] DataType = new object[nRefCount - 1];
                int[] Status = new int[nRefCount - 1];
                object[] RefEntity = new object[nRefCount - 1];
                object[] FeatComp = new object[nRefCount - 1];

                ModelPathName = (object[])vModelPathName;
                ComponentPathName = (object[])vComponentPathName;
                Feature = (object[])vFeature;
                DataType = (object[])vDataType;
                Status = (int[])vStatus;
                RefEntity = (object[])vRefEntity;
                FeatComp = (object[])vFeatComp;
            }
        }
    }
}

```

```
StringBuilder sb = new StringBuilder();

    for (i = 0; i <= nRefCount - 1; i++)
    {
        sb = sb.Append("=>"); sb = sb.Append("Reference Count = "); sb =
sb.Append(nRefCount.ToString()); sb = sb.Append("\r\n");
        sb = sb.Append(" "); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Model path + name = "); sb =
sb.Append(ModelPathName[i].ToString()); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Component path + name = ");
sb = sb.Append(ComponentPathName[i].ToString()); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Feature = "); sb =
sb.Append(Feature[i].ToString()); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Data type = "); sb =
sb.Append(DataType[i].ToString()); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Status = "); sb =
sb.Append(System.Convert.ToString(Status[i])); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Reference entity = "); sb =
sb.Append(RefEntity[i].ToString()); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Feature component = "); sb =
sb.Append(FeatComp[i].ToString()); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Config option = "); sb =
sb.Append(nConfigOpt.ToString()); sb = sb.Append("\r\n");
        sb = sb.Append("=>"); sb = sb.Append("Config name = "); sb =
sb.Append(sConfigName); sb = sb.Append("\r\n");

        SetRichTextBoxVal(sb.ToString());
    }
}
}
```